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$$\begin{aligned} & \lim_{x \rightarrow \infty} \left(\frac{x^2 - 2}{x^2 - 1} \right)^{2x^2 + x + 1} \\ &= \lim_{x \rightarrow \infty} \left(\frac{x^2 - 1 - 1}{x^2 - 1} \right)^{2x^2 + x + 1} \\ &= \lim_{x \rightarrow \infty} \left(1 - \frac{1}{x^2 - 1} \right)^{2x^2 + x + 1} \\ &= \lim_{x \rightarrow \infty} \left(\left(1 + \frac{-1}{x^2 - 1} \right)^{\frac{x^2 - 1}{-1} \cdot \frac{-1}{x^2 - 1}} \right)^{2x^2 + x + 1} \\ &= e^{\lim_{x \rightarrow \infty} \frac{-(2x^2 + x + 1)}{x^2 - 1}} \\ &= e^{\lim_{x \rightarrow \infty} \frac{-2x^2}{x^2}} \\ &= e^{-2} \end{aligned}$$

References